

University of Minnesota
School of Physics and Astronomy

2026 Fall Mathematics 8302
Differential Topology
Assignment Solution

Lecture Instructor: Professor Erkao Bao

Zong-En Chen
chen9613@umn.edu

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Question 1.7

Let N denote the **north pole** $(0, \dots, 0, 1) \in \mathbb{S}^n \subseteq \mathbb{R}^{n+1}$, and let S denote the **south pole** $(0, \dots, 0, -1)$. Define the **stereographic projection** $\sigma : \mathbb{S}^n \setminus \{N\} \rightarrow \mathbb{R}^n$ by

$$\sigma(x_1, \dots, x_{n+1}) = \frac{(x_1, \dots, x_n)}{1 - x_{n+1}}. \quad (1)$$

Let $\tilde{\sigma}(x) = -\sigma(-x)$ for $x \in \mathbb{S}^n \setminus \{S\}$.

- (a) For any $x \in \mathbb{S}^n \setminus \{N\}$, show that $\sigma(x) = u$, where $(u, 0)$ is the point where the line through N and x intersects the linear subspace where $x_{n+1} = 0$. Similarly, show that $\tilde{\sigma}(x)$ is the point where the line through S and x intersects the same subspace. (For this reason, $\tilde{\sigma}$ is also called **stereographic projection from the south pole**.)
- (b) Show that σ is bijective and

$$\sigma^{-1}(u^1, \dots, u^n) = \frac{(2u^1, \dots, 2u^n, |u|^2 - 1)}{|u|^2 + 1}. \quad (2)$$

- (c) Compute the transition map $\tilde{\sigma} \circ \sigma^{-1}$ and verify that the atlas consisting of the two charts $(\mathbb{S}^n \setminus \{N\}, \sigma)$ and $(\mathbb{S}^n \setminus \{S\}, \tilde{\sigma})$ defines a smooth structure on $\mathbb{S}^n$. (The coordinates defined by σ and $\tilde{\sigma}$ are called **stereographic coordinates**.)
- (d) Show that this smooth structure is the same as the one defined in Example 1.31.

Example 1.31 We put a smooth structure on $\mathbb{S}^n$ as follows. For each $i = 1, \dots, n+1$, let $(U_i^\pm, \phi_i^\pm)$ denote the graph coordinate charts, where

$$U_i^\pm = \{(x_1, \dots, x_{n+1}) \in \mathbb{S}^n : \pm x_i > 0\}, \quad \phi_i^\pm(x_1, \dots, x_{n+1}) = (x_1, \dots, \hat{x}_i, \dots, x_{n+1}), \quad (3)$$

where the hat indicates that the i -th coordinate is omitted.

For any distinct indices i, j , the transition map $\phi_i^\pm \circ (\phi_j^\pm)^{-1}$ is easily computed. In the case $i < j$, we get

$$\phi_i^\pm \circ (\phi_j^\pm)^{-1}(u^1, \dots, u^n) = (u^1, \dots, \hat{u}^i, \dots, \pm \sqrt{1 - |u|^2}, \dots, u^n) \quad (4)$$

with the $\pm \sqrt{1 - |u|^2}$ in the j -th position, and a similar formula holds for $i > j$. In the case $i = j$, an even simpler formula holds:

$$\phi_i^\pm \circ (\phi_i^\mp)^{-1}(u^1, \dots, u^n) = \text{ID}_{\mathbb{B}^n}. \quad (5)$$

Thus, the collection of charts $\{(U_i^\pm, \phi_i^\pm)\}$ is a smooth atlas, and so defines a smooth structure on $\mathbb{S}^n$.

Answer

(a)

Consider the line

$$\ell(t) = N + t(x - N) = t(x_1, \dots, x_n, x_{n+1} - 1) + (0, \dots, 0, 1) = (tx_1, \dots, tx_n, t(x_{n+1} - 1) + 1). \quad (6)$$

Obviously, $\ell(0) = N$ and $\ell(1) = x$. We want to find the point where this line intersects the hyperplane $x_{n+1} = 0$. Setting the last coordinate of $\ell(t)$ equal to zero, we have

$$t(x_{n+1} - 1) + 1 = 0 \implies t = \frac{1}{1 - x_{n+1}}. \quad (7)$$

Plugging this value of t into the first n coordinates of $\ell(t)$, we get

$$\ell\left(t = \frac{1}{1 - x_{n+1}}\right) = \frac{(x_1, \dots, x_n)}{1 - x_{n+1}}. \quad (8)$$

This is exactly the definition of $\sigma(x)$. A similar argument shows that $\tilde{\sigma}(x)$ is the point where the line through S and x intersects the hyperplane $x_{n+1} = 0$. More explicitly, consider the line

$$\tilde{\ell}(t) = S + t(x - S) = t(x_1, \dots, x_n, x_{n+1} + 1) + (0, \dots, 0, -1) = (tx_1, \dots, tx_n, t(x_{n+1} + 1) - 1). \quad (9)$$

Setting the last coordinate of $\tilde{\ell}(t)$ equal to zero, we have

$$t(x_{n+1} + 1) - 1 = 0 \implies t = \frac{1}{1 + x_{n+1}}. \quad (10)$$

Plugging this value of t into the first n coordinates of $\tilde{\ell}(t)$, we get

$$\tilde{\ell}\left(t = \frac{1}{1 + x_{n+1}}\right) = \frac{(x_1, \dots, x_n)}{1 + x_{n+1}}. \quad (11)$$

This is exactly the definition of $\tilde{\sigma}(x)$.

(b)

In order to show that σ is bijective, we will show that it is both injective and surjective. For injectivity, suppose that $\sigma(x) = \sigma(y)$ for some $x, y \in \mathbb{S}^n \setminus \{N\}$. Then we have

$$\frac{(x_1, \dots, x_n)}{1 - x_{n+1}} = \frac{(y_1, \dots, y_n)}{1 - y_{n+1}}. \quad (12)$$

Hence, we have

$$x_i = \frac{1 - x_{n+1}}{1 - y_{n+1}} y_i \quad \text{for all } i = 1, \dots, n. \quad (13)$$

Consider the last coordinate of x and y . Since both x and y are on the unit sphere, we have

$$1 - x_{n+1}^2 = \sum_{i=1}^n x_i^2 = \sum_{i=1}^n \left(\frac{1 - x_{n+1}}{1 - y_{n+1}} y_i \right)^2 = \left(\frac{1 - x_{n+1}}{1 - y_{n+1}} \right)^2 \sum_{i=1}^n y_i^2 = \left(\frac{1 - x_{n+1}}{1 - y_{n+1}} \right)^2 (1 - y_{n+1}^2). \quad (14)$$

This implies that

$$1 - x_{n+1}^2 = \left(\frac{1 - x_{n+1}}{1 - y_{n+1}} \right)^2 (1 - y_{n+1}^2) \quad (15)$$

$$\implies (1 - x_{n+1})(1 + x_{n+1}) = \frac{(1 - x_{n+1})^2}{(1 - y_{n+1})^2} (1 - y_{n+1})(1 + y_{n+1}) \quad (16)$$

$$\implies (1 + x_{n+1})(1 - y_{n+1}) = (1 - x_{n+1})(1 + y_{n+1}) \quad (17)$$

$$\implies 1 + x_{n+1} - y_{n+1} - x_{n+1}y_{n+1} = 1 - x_{n+1} + y_{n+1} - x_{n+1}y_{n+1} \quad (18)$$

$$\implies 2x_{n+1} = 2y_{n+1} \implies x_{n+1} = y_{n+1}. \quad (19)$$

If $x_{n+1} = y_{n+1}$, then we have

$$x_i = \frac{1 - x_{n+1}}{1 - y_{n+1}} y_i = y_i \quad \text{for all } i = 1, \dots, n. \quad (20)$$

Thus, we have shown that $x = y$, and hence σ is injective.

Next, in order to show that σ is surjective, let $u \in \mathbb{R}^n$ be arbitrary.

Question 1.9

Complex projective n -space, denoted by $\mathbb{CP}^n$, is the set of all 1-dimensional complex-linear subspaces of $\mathbb{C}^{n+1}$, with the quotient topology inherited from the natural projection $\pi : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{CP}^n$. Show that $\mathbb{CP}^n$ is a compact $2n$ -dimensional topological manifold, and show how to give it a smooth structure analogous to the one we constructed for $\mathbb{RP}^n$. (We use the correspondence

$$(x^1 + iy^1, \dots, x^{n+1} + iy^{n+1}) \longleftrightarrow (x^1, \dots, x^{n+1}, y^1, \dots, y^{n+1}) \quad (21)$$

to identify $\mathbb{C}^{n+1}$ with $\mathbb{R}^{2n+2}$.)

Answer